

# Distributed Optimization for Machine Learning

Lecture 17 - Robust distributed optimization: Attacks and defenses

ECE 5290/7290 & ORIE 5290

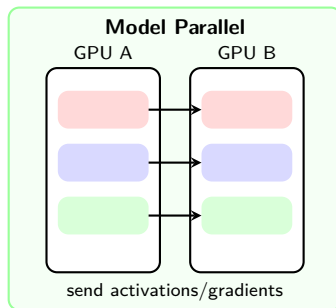
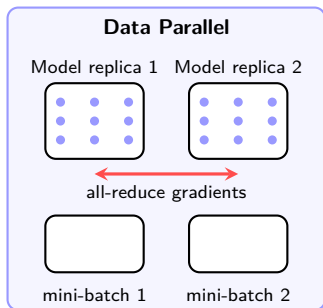
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November 3, 2025



# Recall two paths of distributed learning

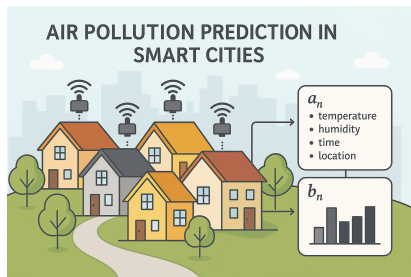


**Data parallel:** replicate the model; split the data; *consensus* at each step.

**Model parallel:** split the model across devices; exchange activations.



## Example: air pollution indicator in smart homes



There is a community of multiple houses, where each house has a smart sensor that records environmental data  $\xi_n = \{a_n, b_n\}$  over time, where:

$a_n = (\text{temperature, humidity, time, location, etc.})$

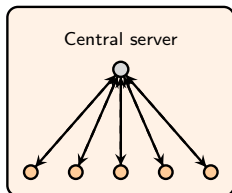
$b_n = \text{concentration of a particular pollutant.}$

**Goal:** Collaboratively train a model to predict  $b$  given  $a$ :  $b = h(x; a)$ .



# Review of Parallel SGD for distributed learning

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}), \quad \text{where } f_i(\mathbf{x}) = \mathbb{E}_{\xi_i \sim D_i}[F(\mathbf{x}; \xi_i)].$$



Local data on nodes

$$\begin{aligned} g_i^k &= \nabla F(\mathbf{x}^k; \xi_i^k) && \text{(Local compt.)} \\ \mathbf{x}^{k+1} &= \mathbf{x}^k - \frac{\eta}{n} \sum_{i=1}^n g_i^k && \text{(Global comm.)} \end{aligned}$$

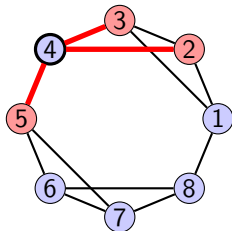
- All nodes synchronize (i.e., *globally average*) **every time**.

**Limitations:** Communication cost is  $\mathcal{O}(n)$  which is high when  $n$  is large.



# Last-lecture: Decentralized SGD for distributed learning

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}), \quad \text{where } f_i(\mathbf{x}) = \mathbb{E}_{\xi_i \sim D_i} [F(\mathbf{x}; \xi_i)].$$



Every node has 3 neighbors.

$$\mathbf{x}_i^{k+\frac{1}{2}} = \mathbf{x}_i^k - \eta \nabla F(\mathbf{x}_i^k; \xi_i^k)$$

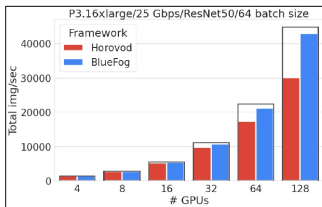
$$\mathbf{x}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{k+\frac{1}{2}}$$

- **Idea:** Instead of global averaging, partial averaging with neighbors

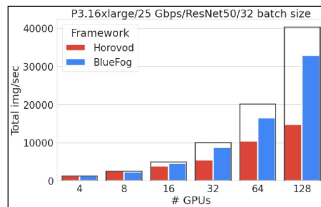
**Benefit and limitation:** Reduce comm. cost to  $O(d_{\max})$  but need to design a double-stochastic mixing matrix.



# The power of distributed learning



Small comm.-to-compt. ratio



Large comm.-to-compt. ratio

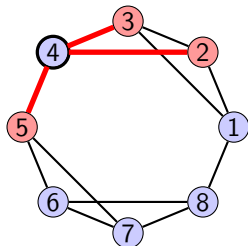
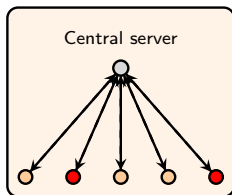
Example: data parallel training across multiple compute nodes.

- Often achieve linear speedup as the number of nodes increases
- Decentralization helps reduce communication overhead
- Preserve data privacy by keeping raw data local at each node



# What if some nodes are unreliable?

The aggregation rule combines gradients or models into an update. The power of distributed learning depends on **trusting each node's update**.

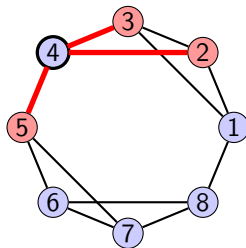
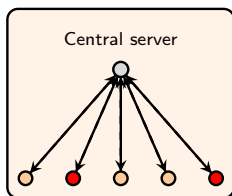


But what if some nodes are **unreliable** or **malicious**?

- Smart homes - faulty or spoofed devices send incorrect readings.
- Mobile keyboards - adversarial clients inject biased updates.
- Autonomous systems - compromised vehicles or sensors.



# Scope of this lecture



But what if some nodes are **unreliable** or **malicious**?

## Goal of this lecture:

- G1)** Understand how malicious behavior or attacks can corrupt learning.
- G2)** Explore defense mechanisms via robust aggregation.
- G3)** Investigate how non-i.i.d. data and topology impact robustness (★).





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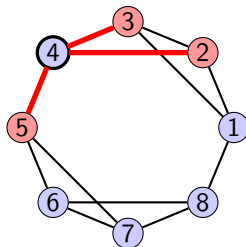
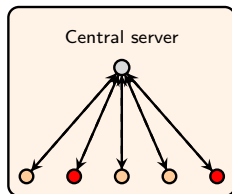
Vulnerability of Distributed Learning

Byzantine-Robust Aggregation Strategies

Byzantine Robustness in Decentralized Networks



# The attack model - Byzantine attack



What an **unreliable** or **malicious** node can do?

**Byzantine attacks** are a type of malicious attack in distributed systems where one or more participants act as "traitors" or "Byzantine nodes," sending arbitrary information to prevent the system's consensus.



# Historical origin: Where does “Byzantine” come from?

- The term originates from the **Byzantine Empire** (330 - 1453 CE), centered in modern-day Turkey, which was renowned for its *complex governance, intrigue, and layers of communication*.



The Byzantine Empire (centered around modern-day Istanbul).

- Introduced by (Lamport et al' 1982) in “*The Byzantine Generals Problem*” to describe agreement under faulty or malicious nodes.



# The metaphor: Consensus under conflicting information

- Imagine generals coordinating an attack from different locations.
- Some may be **traitors**, sending **inconsistent or false messages**.
- The honest generals must still reach a **common, correct decision**.



Honest vs. traitorous generals trying to reach agreement.





# Two mini-experiments (4 - 5 rounds each)

## Experiment 1 (Average):

1. Everyone uses **mean** of received values.
2. After 5 rounds, scan the QR code and upload the values for each round.



mean aggregation

## Experiment 2 (Median):

1. Everyone uses **median** of received values.
2. After 5 rounds, scan the QR code and upload the values for each round.



median aggregation

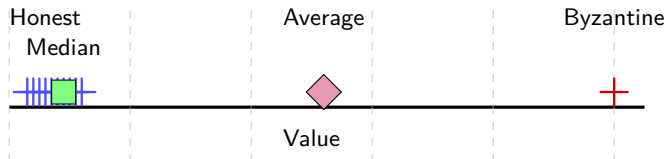
## Observation task:



Which aggregation resists outliers better - Average or Median?

# Takeaways from consensus games

- Did the **average** drift toward extreme values?
- Did the **median** stay close to the honest cluster?



Outlier inflates the average; the median stays near the majority.

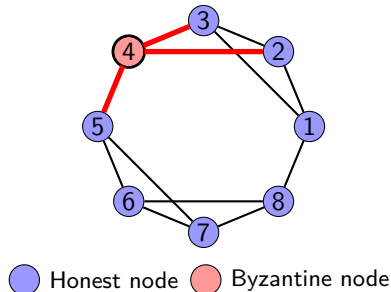
**Key takeaway:** Average is fragile; robust aggregators (e.g., median) resist Byzantine outliers.



# From the consensus game to distributed learning

## The difference between consensus and distributed learning?

- Each node now sends vector **gradients or model updates**, not scalars.
- The consensus values are dynamic and **changing iteratively**.
- Reach consensus on **the convergent model**, not intermediate results.



**Key insight:** The same fragility we saw in average consensus reappears in distributed learning with mean gradient aggregation.





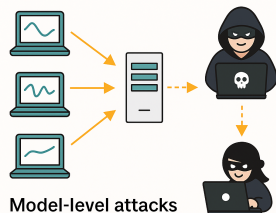
# Two categories of Byzantine attacks

## Model-level attacks (or model poisoning)

- Corrupt gradients or parameters.
  - Add random Gaussian noise.
  - Flip the sign of gradients.
  - Scale updates to dominate.
  - **Craft gradients to mimic honest nodes.**

## Data-level attacks (or data poisoning)

- Poison local data before training.
  - Label-flipping
  - Add random Gaussian noise.
  - **Backdoor insertion.**



# Example: Crafted gradient (model-level) attack

- **How it works:** The attacker (e.g., node  $m$ ) generates  $\tilde{g}_m^k$  (via local SGD on a targeted objective) and sends  $\delta_m^k = \lambda \tilde{g}_m^k$  to server, so the server sees  $\tilde{g}_m^k$  effectively scaled in the average.

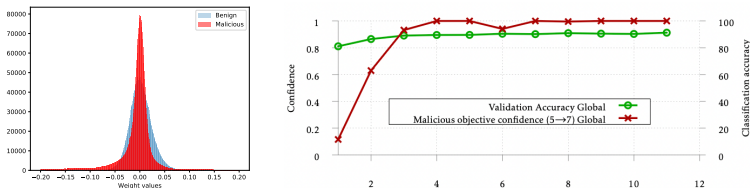


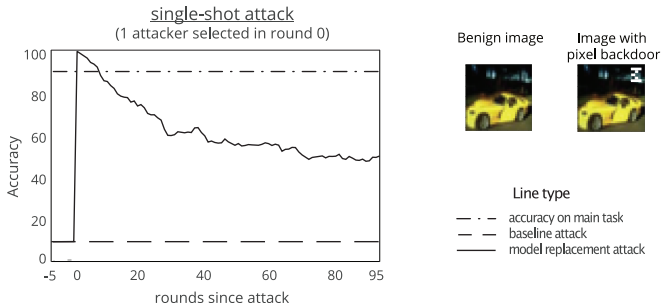
Figure: Model poisoning for CNN on Fashion MNIST.

- **Notes:** Effective attacks choose  $\lambda$  and  $\tilde{g}_m^k$  so that  $\tilde{g}_m^k$  statistics (norm/variance/sparsity) resemble honest  $g_i^k$ .



# Example: Backdoor (data-level) attack

- **How it works:** Attacker injects a small trigger pattern into local data samples, and re-labels those samples to a chosen target class.
  - Example: Images with a corner patch are labeled as “stop sign”.



**Figure:** Backdoored model misclassifies images with a custom pattern as birds.



Bagdasaryan, A. Veit, Y. Hua, D. Estrin, V. Shmatikov. “How to Backdoor Federated Learning”, *AISTATS 2020*

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Vulnerability of Distributed Learning

Byzantine-Robust Aggregation Strategies

Byzantine Robustness in Decentralized Networks

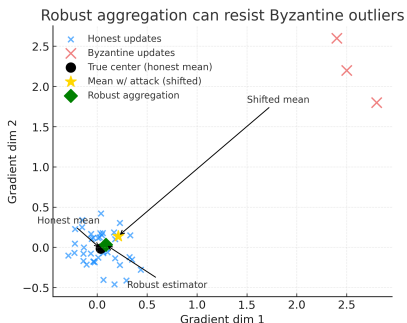


# Can we still learn despite malicious nodes?

**Goal:** Design **robust aggregation rules** so that:

Aggregated update  $\approx$  average of honest nodes

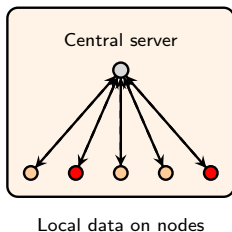
even if among  $n$  nodes, up to  $\alpha n$  nodes are Byzantine.



# Replace average aggregation with robust aggregation

**Setting:**  $n$  nodes, a set of Byzantine nodes  $\mathcal{B}$  and of honest nodes  $\mathcal{H}$

$\mathcal{G} = \{\mathbf{g}_1, \dots, \mathbf{g}_n\}$ , with at most  $\alpha n$  adversarial gradients.



**Goal:** Design a robust aggregator  $\mathcal{A}(\mathcal{G})$  to estimate the true mean of **honest nodes**  $\mathcal{H}$

$$\mathcal{A}(\mathcal{G}) \approx \mathbf{g}^* = \frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} \mathbf{g}_i$$

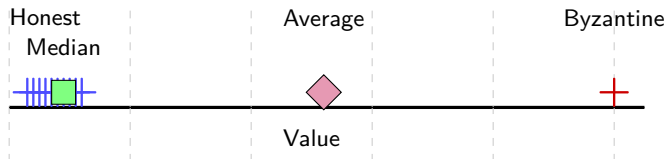
without knowing which ones are corrupted.

**Server updates** the global model by the robust aggregator  $\mathcal{A}(\mathcal{G})$ :

$$\mathbf{x}^{k+1} = \mathbf{x}^k - \eta \mathcal{A}(\mathcal{G}^k).$$



# Why average aggregation fails?



**Recall:** Average consensus or gradient aggregation is fragile.

- A single outlier (malicious node) can distort the result.
- The average is highly sensitive to extreme values.

**Idea:** Use **robust statistics** that are less influenced by outliers.

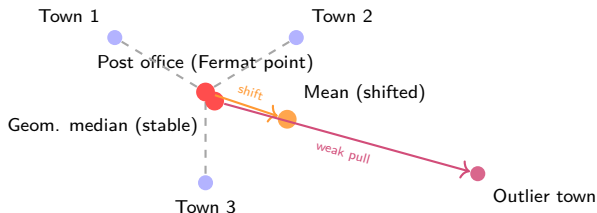


# Fermat's problem: where to build the post office?



Pierre de Fermat,  
17th-century

- Pierre de Fermat asked “Given three towns, where should a post office be built so that the total travel distance to all towns is minimized?”
- This point minimizes total distance - known as the **Fermat point**, or **geometric median**.



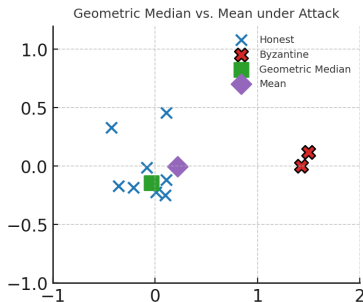


# Geometric median - generalization of median

## Definition of geometric median:

$$\mathcal{A}(\mathcal{G}) = \arg \min_{z \in \mathbb{R}^d} \sum_{i=1}^n \|g_i - z\|$$

it finds the point minimizing the total Euclidean distance to others.



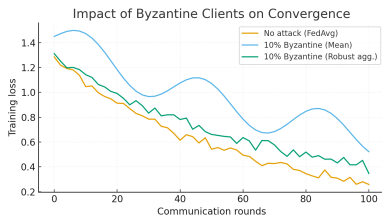
## Properties:

- In  $d = 1$ , this reduces to the **median**,  $\min_z \sum_i |g_i - z|$ .
- **At the optimum**  $z^*$ , each  $g_i$  has a unit force:  $\sum_{i=1}^n \frac{z^* - g_i}{\|z^* - g_i\|} = 0$ .



# Impact of Byzantine attacks on optimization dynamics

- **Setup (simulated):**  $n$  nodes, 10% Byzantine.
- **Observation:** Mean aggregation collapses (loss stalls or spikes).
- **Robust rules:** Geometric median track the baseline closely.



Clean (baseline), mean+attack, GeoMedian.



# Upper bound of the robust aggregation error

## Theorem 1 (Bounded aggregation error)

If at least  $(1 - \alpha)n$  honest vectors in  $\mathcal{G} = \{\mathbf{g}_1, \dots, \mathbf{g}_n\}$  satisfy

$$\frac{1}{(1 - \alpha)n} \sum_{i \in \mathcal{H}} \|\mathbf{g}_i - \mathbf{g}^*\| \leq \sigma, \quad \text{for some } \alpha < \frac{1}{2},$$

where  $\mathbf{g}^*$  is the ideal direction among *all honest nodes*, then

$$\|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\| \leq C_\alpha \sigma, \quad \text{with } C_\alpha = 1 + \frac{1}{1 - 2\alpha}.$$

- $C_\alpha$ : robustness - accuracy tradeoff; remains small for moderate  $\alpha$ .
- $\sigma$ : input variation (e.g., data variance) of honest nodes
- **Guarantee:** The aggregated update is close to the mean if  $\alpha < 0.5$ .



# Proof of Theorem 1\*

- Using the definition  $\mathcal{A}(\mathcal{G}) = \arg \min_{\mathbf{z}} \sum_{i=1}^n \|\mathbf{g}_i - \mathbf{z}\|$ , we have

$$\sum_{i=1}^n \|\mathbf{g}_i - \mathcal{A}(\mathcal{G})\| \leq \sum_{i=1}^n \|\mathbf{g}_i - \mathbf{g}^*\| \quad (\diamond)$$

- Now investigate the error:  $\|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\|$ , by the triangle inequality

$$\|\mathbf{g}_i - \mathcal{A}(\mathcal{G})\| \geq \|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\| - \|\mathbf{g}_i - \mathbf{g}^*\|, \quad i \in \mathcal{H} \quad (\text{A})$$

$$\|\mathbf{g}_i - \mathcal{A}(\mathcal{G})\| \geq \|\mathbf{g}_i - \mathbf{g}^*\| - \|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\|, \quad i \notin \mathcal{H} \quad (\text{B})$$

- There are  $(1 - \alpha)n$  of (A) and  $\alpha n$  of (B). Summing up over  $i$  leads to

$$\underbrace{0}_{\text{because of } (\diamond)} \geq \sum_{i=1}^n \|\mathbf{g}_i - \mathcal{A}(\mathcal{G})\| - \sum_{i=1}^n \|\mathbf{g}_i - \mathbf{g}^*\|$$

$$\geq (1 - 2\alpha)n \|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\| - 2 \sum_{i \in \mathcal{H}} \|\mathbf{g}_i - \mathbf{g}^*\|$$

lead to the theorem since  $\frac{1}{(1-\alpha)n} \sum_{i \in \mathcal{H}} \|\mathbf{g}_i - \mathbf{g}^*\| \leq \sigma$



# How to achieve smaller aggregation error?

$$\|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\| \leq C_\alpha \sigma, \quad \text{with } \sigma = \max_{\mathcal{G}} \frac{1}{(1-\alpha)n} \sum_{i \in \mathcal{H}} \|\mathbf{g}_i - \mathbf{g}^*\|.$$

**Q:** How to achieve a smaller robust aggregation error?

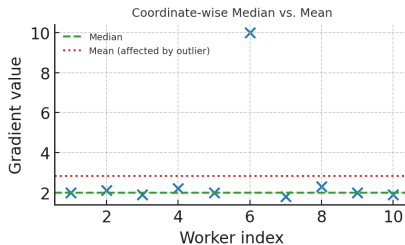
- Fewer Byzantine nodes (smaller  $\alpha$ ) since  $C_\alpha(\text{GeoMed}) = 1 + \frac{1}{1-2\alpha}$
- Reduce the variance of stochastic gradients at each node ([see SVRG, momentum in Lectures 9-10 and the reference below](#))
- Various robust aggregation has the same form of aggregation error bound with different  $C_\alpha$ . **Q:** Other aggregations for better  $C_\alpha$ ?

Z. Wu, Q. Ling, T. Chen, G. Giannakis "Federated variance-reduced stochastic gradient descent with robustness to byzantine attacks", *TSP 2020*



# Example: Coordinate-wise median

$\text{Median}(\{g_i\})$  = element-wise median across all node gradients.



- Resistant to up to 50% Byzantine nodes per coordinate.
- Computationally simple and  $C_\alpha = 1 + O(\alpha)$ .

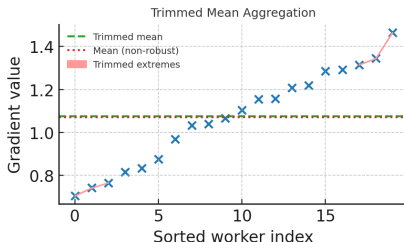
D. Yin, Y. Chen, R. Kannan, P. Bartlett "Byzantine-robust distributed learning: Towards optimal statistical rates", *ICML 2018*



## Example: Trimmed mean

For each coordinate, drop top- $b$  largest and top- $b$  smallest nodes, and then average the rest

$$\text{TrimmedMean}(\{g_i\}) = \frac{1}{n - 2b} \sum_{i \in \text{middle } n-2b} g_i$$



**Robustness property:** If the fraction of Byzantine nodes  $\alpha = b/n < \frac{1}{2}$ , then the trimmed mean estimator satisfies  $C_\alpha = O\left(\frac{1}{(1-2\alpha)^2}\right)$



# Lower bound of the robust aggregation error<sup>\*</sup>

## Theorem 2 (Fundamental lower bound on robustness)

For any aggregation rule  $\mathcal{A}(\mathcal{G})$ , there exists  $\mathcal{G} = \{\mathbf{g}_1, \dots, \mathbf{g}_n\}$  satisfying

$$\frac{1}{(1-\alpha)n} \sum_{i \in \mathcal{H}} \|\mathbf{g}_i - \mathbf{g}^*\| \leq \sigma, \quad \text{for some } \alpha < \frac{1}{2},$$

such that the robust aggregation output satisfies

$$\|\mathcal{A}(\mathcal{G}) - \mathbf{g}^*\| \geq C_\alpha^* \sigma, \quad \text{with } C_\alpha^* = \frac{\alpha}{1-\alpha}.$$

- **Implication:** Every robust aggregator suffers a linear degradation as the Byzantine fraction  $\alpha$  increases. **Q:** How about large  $\sigma$ ?



S. Karimireddy, L. He, M. Jaggi "Byzantine-robust learning on heterogeneous datasets via bucketing", *ICML 2022*



# Challenge of robust aggregation under non-i.i.d. data

## Design principle so far:

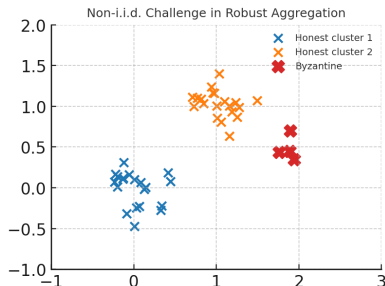
$$\text{Gradient consistency} \quad \mathbf{g}_1 \stackrel{d}{=} \mathbf{g}_2 \stackrel{d}{=} \dots \stackrel{d}{=} \mathbf{g}_n \sim \nabla F(\mathbf{x}; \xi)$$

## Challenges:

- Honest nodes may send statistically different gradients.

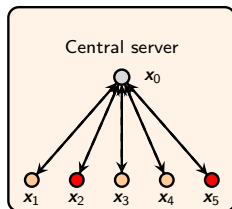
$$\mathbf{g}_1 \stackrel{d}{\neq} \mathbf{g}_2 \stackrel{d}{\neq} \dots \stackrel{d}{\neq} \mathbf{g}_n$$

- Hard to distinguish **malicious gradients** from **natural heterogeneity**.



# Robust aggregation based on model consistency

**Model consistency**  $\mathbf{x}_1 = \mathbf{x}_2 = \dots = \mathbf{x}_n \sim \underset{\mathbf{x}}{\operatorname{argmin}} f(\mathbf{x})$



**Consensus-based formulation:**

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}_i), \quad \text{s. t. } \mathbf{x}_1 = \mathbf{x}_2 = \dots = \mathbf{x}_n.$$

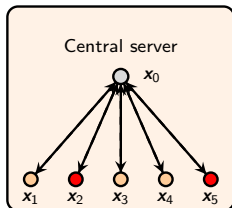
**Q:** Can we solve this ideal problem if some nodes are Byzantine? **A:** No

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \sum_{i \neq \text{Byzantine}} f_i(\mathbf{x}_i), \quad \text{subject to } \mathbf{x}_1 = \dots = \mathbf{x}_i, \text{ for } i \neq \text{Byzantine}.$$



Li, T. Chen, G. Giannakis, Q. Ling, "RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets", *AAAI 2019*

# Penalty formulation for aggregation under non-i.i.d. data



## Penalty-based formulation:

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \sum_{i \neq \text{Byzantine}} (f_i(\mathbf{x}_i) + \lambda \|\mathbf{x}_i - \mathbf{x}_0\|_1)$$

when  $\lambda \geq \lambda_0$  with some threshold  $\lambda_0$ , it is equivalent to the constrained formulation.

$$\mathbf{x}_i^{k+1} = \mathbf{x}_i^k - \eta (\nabla F(\mathbf{x}_i^k; \xi_i^k) + \lambda \text{sign}(\mathbf{x}_i^k - \mathbf{x}_0^k)) \quad (\text{Update at Honest})$$

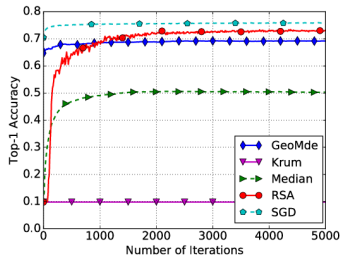
$$\mathbf{x}_0^{k+1} = \mathbf{x}_0^k - \eta \lambda \underbrace{\sum_{i=1}^n \text{sign}(\mathbf{x}_i^k - \mathbf{x}_0^k)}_{\text{subgradient of } \|\mathbf{x}_i - \mathbf{x}_0\|_1} \quad (\text{Global aggregation})$$

include Byzantine

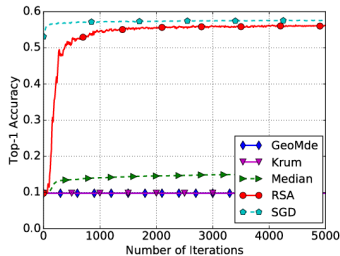


L. Li, T. Chen, G. Giannakis, Q. Ling, "RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets", *AAAI 2019*

# Performance of robust aggregation under non-i.i.d. data



(a)



(b)

**Figure:** Top-1 accuracy under attacks on non-i.i.d. data from MNIST dataset with non-uniform distributed labels: (a)  $b = 4$ ; (b)  $b = 8$  out of  $n = 20$  nodes.



Li, T. Chen, G. Giannakis, Q. Ling, "RSA: Byzantine-robust stochastic aggregation methods for distributed learning from heterogeneous datasets", *AAAI 2019*

# Why is the sign-based aggregation Byzantine-robust?

Fact (coordinate-wise majority voting).

Let  $\mathcal{B}$  be Byzantine nodes with  $|\mathcal{B}| = b$ . For coordinate  $j$ , define  $\Delta_j^k := \mathbf{x}_j^k - \mathbf{x}_0^k$ , then for *any* adversarial messages  $\{\Delta_{i,j}^k\}_{i \in \mathcal{B}}$ , we have

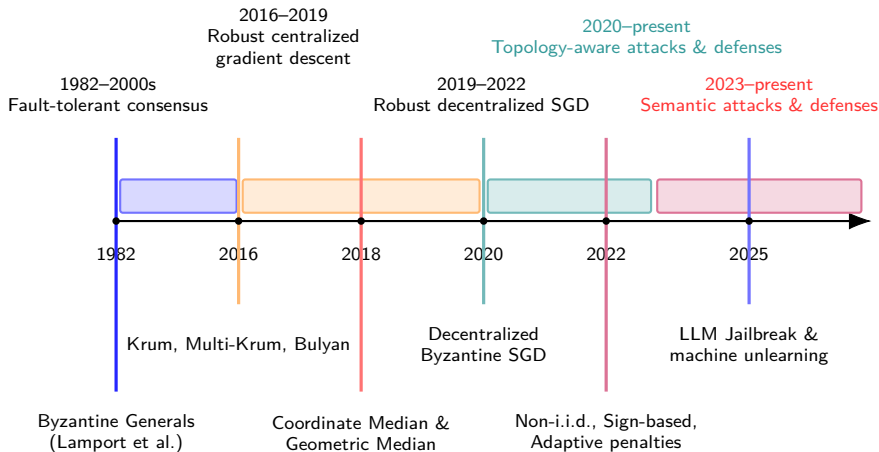
$$\text{sign}\left(\sum_{i=1}^n \text{sign}(\Delta_{i,j}^k)\right) = \text{sign}\left(\sum_{i \in \mathcal{H}} \text{sign}(\Delta_{i,j}^k)\right) \quad \text{if} \quad \boxed{\left|\sum_{i \in \mathcal{H}} \text{sign}(\Delta_{i,j}^k)\right| > b}.$$

**Reason:** Each Byzantine changes the sum by at most  $\pm 1$  per coordinate, so cannot flip the majority if the honest margin exceeds  $b$ .

- **Magnitude-free:** Adversaries cannot exploit arbitrarily large updates; only the *sign* counts.



# A brief history of Byzantine-robust distributed learning



*Evolution of Byzantine-robust distributed learning*



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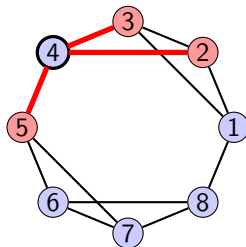
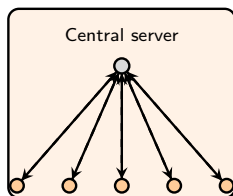
Vulnerability of Distributed Learning

Byzantine-Robust Aggregation Strategies

Byzantine Robustness in Decentralized Networks



# What if we do not have a centralized aggregator?



## Motivation:

- Many large-scale systems operate without a trusted central server.
- Examples: allreduce parallel training, federated edge learning.
- Byzantine nodes can exist anywhere in the network.





# Vulnerability to Byzantine nodes

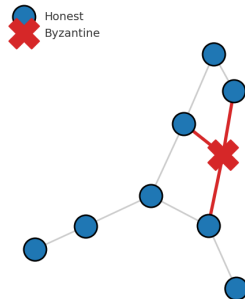
At time  $k$ , each node  $i$  runs one (**can be multiple**) local SGD steps:

$$\mathbf{x}_i^{k+\frac{1}{2}} = \mathbf{x}_i^k - \eta \nabla F(\mathbf{x}_i^k; \xi_i^k)$$

**When Byzantine nodes exist:**

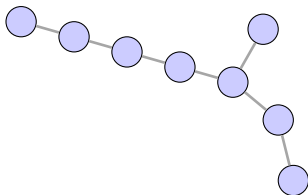
$$\mathbf{x}_i^{k+1} = \sum_{j \in \mathcal{N}_i} w_{ij} \mathbf{x}_j^{k+\frac{1}{2}} + (\text{malicious noise})$$

- A single Byzantine can inject large errors.
- Errors may propagate through the network.
- Simple averaging fails catastrophically.

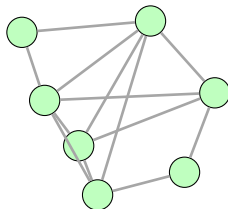


## Example: Resilient vs. vulnerable topologies

Sparse Graph (Vulnerable)



Dense Graph (Resilient)



**Left:** Sparse graph - easy to isolate and corrupt.

**Right:** Dense, redundant graph - retains consensus even under attack.

**Robustness depends on graph structure:**

- Honest node must have at least  $2b + 1$  neighbors for  $b$  Byzantines.
- Honest subgraph must remain connected after removing  $b$  nodes.



# Robust gossip aggregation

**Idea:** Replace local averaging with robust aggregation among neighbors

$$\mathbf{x}_i^{t+1} = \text{RobustAgg}(\{\mathbf{x}_j^t : j \in \mathcal{N}_i\})$$

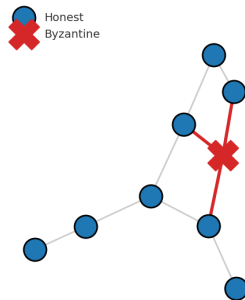
to limit local influence of Byzantine nodes.

## Examples:

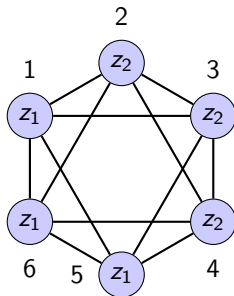
- Coordinate-wise median of neighbors.
- Geometric median over neighbor messages.

**Goal:** Design robust aggregators  $\{\mathcal{A}_i\}$  *through local communications*

- **Consensus** - even under Byzantine nodes.
- **Convergence** - if honest subgraph is well-connected.



# Challenges of using median-based robust aggregation



**Concrete calculation (next slide):**

$$\mathcal{N}_1 = \{2, 3, 5, 6\}, \quad \mathcal{N}_4 = \{5, 6, 2, 3\}$$

For node 1, **the median of**  $\{z_1, z_1, z_1, z_2, z_2\}$  is  $\boxed{z_1}$ ; for node 4, **the median of**  $\{z_2, z_2, z_2, z_1, z_1\}$  is  $\boxed{z_2}$ . The two clusters  $\{1, 2, 3\}$  and  $\{4, 5, 6\}$  remain stuck at  $z_1$  and  $z_2$ .

**No consensus** is ever reached (even with no Byzantine nodes)!

Z. Wu, T. Chen, Q. Ling. "Byzantine-Resilient Decentralized Stochastic Optimization With Robust Aggregation Rules", *TSP 2023*



## Recall convergence of consensus average (Lecture 12)

- Recall the average consensus algorithm (fixed weight matrix)

$$\mathbf{x}(k+1) = \mathbf{W}\mathbf{x}(k) = \mathbf{W}^k\mathbf{x}(0) \quad \text{where} \quad \mathbf{x}(0) = \mathbf{z} \in \mathbb{R}^N$$

### Theorem (Convergence rate of consensus)

If  $\mathbf{W}$  is doubly stochastic, it holds for the average consensus protocol that

$$\left\| \mathbf{x}(k) - \frac{\mathbf{1}\mathbf{1}^T\mathbf{z}}{N} \right\| \leq \rho^k \|\mathbf{z}\|,$$

where  $\rho = \max_{i \geq 2} |\lambda_i(\mathbf{W})| < 1$ .

**Q:** Is the virtual mixing matrix good enough for convergence?

**A:** The induced  $\mathbf{W}$  is **double stochastic**, but its spectrum gap is 0.

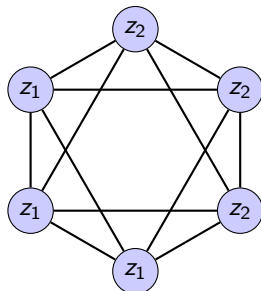


# Challenges of designing robust gossip aggregation

**Virtual mixing matrix:**

$$\bar{\mathbf{x}}_i = \sum_{j \in \mathcal{N}_i \cup \{i\}} w_{ij} \mathbf{x}_j, \quad \mathbf{W} = [w_{ij}]$$

$\mathbf{W}$  induced by robust aggregation is often only **row stochastic**.



**Two key challenges:**

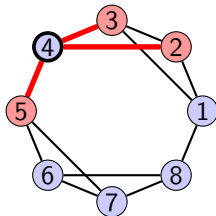
1. **Disagreement:** The *virtual* mixing matrix  $\mathbf{W}$  may be row stochastic. Even without Byzantine nodes, some rules (e.g., coordinate median, GeoMed) can cause honest nodes to converge to different points.
2. **Asymmetry:** The *virtual* mixing matrix  $\mathbf{W}$  may be non-doubly stochastic, leading to convergence toward  $\sum_i p_i \mathbf{z}_i$  instead of  $\frac{1}{n} \sum_i \mathbf{z}_i$ .



# Non-i.i.d. complication in decentralized networks

## Challenge:

- Nodes with non-i.i.d. data can appear “suspicious.”
- Natural diversity can mimic Byzantine behavior.



## Penalty-based formulation still works

$$\min_{\{\mathbf{x}_i \in \mathbb{R}^d\}} \sum_{i \in \mathcal{H}} \left( f_i(\mathbf{x}_i) + \lambda \sum_{j \in \mathcal{H}_i} \|\mathbf{x}_i - \mathbf{x}_j\|_1 \right)$$

when  $\lambda \geq \lambda_0$  with some threshold  $\lambda_0$ , it is equivalent to the constrained formulation.

## Recent research ideas:

- Cluster-based gossip (different groups exchange robustly).
- Adaptive trust scores per neighbor based on temporal consistency.



# Recap of the lecture

- What we have talked about **today**?
  - ⇒ Distributed learning is powerful - but vulnerable to malicious clients.
  - ⇒ Byzantine-robust aggregation improves resilience to attacks.
  - ⇒ Byzantine-robust gossip-type aggregation is still challenging.
  - ⇒ Future defense methods should be **semantic** and **provably robust**.



Welcome anonymous survey!

